EFFECT OF FACTORS AFFECTING ENZYME ACTIVITY

(e.g. Salivary Amylase)

OBJECTIVE: To determine the effect of factors affecting enzyme activity (using Salivary Amylase)

PRINCIPLE:

Alpha amylase is produced by salivary glands and pancreas. It hydrolyses starch into maltose by acting on 1 – 4 glycosidic bonds yielding a mixture of maltose, glucose and oligosaccharides. The saliva secreted into oral cavity contains electrolytes (Na+, k+, Cl-, HCO3 -) and enzymes, salivary amylase and lysozyme. The chemical process of digestion is initiated in the oral cavity by the hydrolytic action of the carbohydrate splitting enzyme, the salivary amylase. Amylase has an optimum pH of 7 (pH of saliva). When saliva is incubated with starch and tested with iodine, the test color turns blue initially (starch), changes successively to violet, then to pink (maltose), and finally disappears (glucose) as salivary amylase disrupts the starch molecules.

REAGENTS:

- Substrate (5% starch solution weight 5 gram of starch in weighing machine and dissolve in 100 ml of distilled water.
- Amylase Enzyme solution: Saliva – Take a mouthful of water. keep rolling in mouth as long as you can. Then split it out in a beaker (stock solution of Amylase enzyme).
- Iodine solution.

- 1% NaCl: (It is necessary for enzyme activity). Weight 1 gram NaCl in weighing machine and dissolve in 100 ml of distilled water.
- 10% HCl: Take 10ml of HCl and dissolve it in distilled water to make 100ml.

PROCEDURE:

TEST TUBE	5% STARCH SOLUTON	AMYLASE ENZYME SOLUTION	1% NaCl	10% HCl	IODINE SOLUTION	TEMPERATURE	FACTORS AFFECTING ENZYME ACTIVITY
1	1ml	-	-	-	2 – 3 drops	Room temp for 15 minutes	Standard
2	1ml	3ml	0.5ml	-	2-3 drops	Room temp for 15 minutes	Normal Enzyme Activity
3	1ml	3ml	0.5ml	1ml	2-3 drops	Room temp for 15 minutes	Acidic / Base PH
4	1ml	3ml	0.5ml	-	2-3 drops		Heat

OBSERVATION:

TEST TUBE	IODINE TEST	INFERANCE	FACTORS AFFECTING ENZYME ACTIVITY
1	Blue color	Starch is present	Standard
2	Light Pink	Starch has been hydrolyzed to Maltose	Normal Enzyme Activity
3	Blue color	No change in Starch	Acidic pH

4	Blue color	No change occurs	Heat

RESULT:

Enzyme activity is destroyed by extreme changes in temperature and Ph

Reaction time= 15 minutes room temperature.

TEST TUBE	REACTANTS	OBSERVATION
A	Starch + Iodine	
B	Starch + Iodine+ Amylase	
C	Starch + Iodine + Amylase+ 10% HCL	
D	Starch + Iodine + Amylase + Heat	

INTERPRETATION: